

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS:

Question 1

Question:

A company needs to transfer a 150 MB design file... Explain how the Transport and Data Link layers ensure reliable communication.

Solution:

Given:

File=150 MB=150,000,000 bytes

Segment size=1500 bytes

Bandwidth=50 Mbps

Data Link overhead=40 bytes/frame

Step 1: Number of segments = $150,000,000 / 1500 = 100,000$ segments.

Each segment becomes one frame.

Frames transmitted=100,000.

Step 2: Total Data Link overhead

= $100,000 \times 40 = 4,000,000$ bytes = 4 MB.

Step 3: Total data transmitted

= $150 + 4 = 154$ MB.

154 MB=1232 Mb.

Transmission time = $1232 / 50 = 24.64$ s.

Answer:

- Segments=100,000
- Frames=100,000
- Overhead=4 MB
- Transmission time=24.64 s

Explanation:

Transport Layer performs segmentation, sequencing, acknowledgements, error recovery and flow control.

Data Link Layer creates frames, adds MAC addresses and CRC for error detection, ensuring reliable node-to-node delivery.

Question 2

Question:

A Transport Layer protocol uses a sliding window size of 6...

Solution:

Given: Window size=6, Total frames=48.

Windows = $48 / 6 = 8$.

One lost frame/window $\Rightarrow$ Retransmissions=8.

Answer:

Windows required=8

Retransmissions=8

Explanation:

Flow control prevents receiver buffer overflow, improves throughput, avoids congestion and increases communication efficiency.

Question 3

Question:

A 12,000-byte IP packet is transmitted through MTU 1500...

Solution:

Data per fragment=1500-20=1480 bytes.

Fragments=12000/1480=8.11≈9.

Header overhead=9×20=180 bytes.

Answer:

Fragments=9

Header overhead=180 bytes.

Explanation:

Network layer fragments packets and destination reassembles them using fragment offset and identification.

Question 4

Question:

A Transport Layer protocol uses a sliding window size of 6...

Solution:

Given: Window size=6, Total frames=48.

Windows=48/6=8.

One lost frame/window ⇒ Retransmissions=8.

Answer:

Windows required=8

Retransmissions=8

Explanation:

Flow control prevents receiver buffer overflow, improves throughput, avoids congestion and increases communication efficiency.

Question 5

Question:

Multimedia application transfers 600 MB with checkpoints every 60 MB...

Solution:

Failure at 390 MB.

Completed checkpoints=390/60=6.

Last checkpoint=360 MB.

Retransmitted data=390-360=30 MB.

Answer:

Checkpoints=6

Retransmission=30 MB.

Explanation:

Synchronization checkpoints allow transmission to restart from the latest checkpoint instead of the beginning.

Question 6

Question:

Presentation Layer compresses a 900 MB file by 40%...

Solution:

Compressed size=900×60%=540 MB.

Encrypted size=540×1.05=567 MB.

Reduction=(900-567)/900×100=37%.

Answer:

Compressed=540 MB

Encrypted=567 MB

Reduction=37%.

Explanation:

Compression saves bandwidth while encryption provides confidentiality.

Question 7

Question:

FTP transfers 3 GB at 120 Mbps with requests of 3 MB...

Solution:

3 GB=3000 MB.

Requests=3000/3=1000.

Transmission time=(3GB=24000 Mb)/120=200 s.

Answer:

Requests=1000

Time=200 s.

Explanation:

Application layer provides FTP services, authentication and file transfer.

Question 8

Question:

Packet processing delays at different OSI layers...

Solution:

Total delay=4+3+5+2+1+18+12=45 ms.

Answer: 45 ms.

Explanation:

Application-user services, Transport-reliability, Network-routing, Data Link-framing, Physical-bit transmission, plus propagation and transmission delays.

Question 9

Question:

Network transfers 80 MB useful data...

Solution:

Useful/frame=1600-80=1520 bytes.

Frames=80,000,000/1520≈52,633.

Overhead=52,633×80=4,210,640 bytes≈4.21 MB.

Efficiency=(1520/1600)×100=95%.

Answer:

Frames=52,633

Overhead≈4.21 MB

Efficiency=95%.

Explanation:

Protocol overhead reduces efficiency but provides addressing, error detection and control.

Question 10

Question:

Hospital transfers 240 MB medical record...

Solution:

Compressed size=240×70%=168 MB.

Segments=168,000,000/1400=120,000.

Overhead=120,000×60=7,200,000 bytes=7.2 MB.

Final transmitted=168+7.2=175.2 MB.

Answer:

Compressed=168 MB

Segments=120,000

Overhead=7.2 MB

Final data=175.2 MB.

Explanation:

Presentation compresses, Transport segments, Data Link frames with overhead, Physical transmits the bits securely and reliably.